

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	Total(10)
Weightage	20%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I M.Mahathi(192324098)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

M.Mahathi

Signature of the student:

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer Income (₹) Spending Score

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

Q1: K-Means Vs Hierarchical Clustering

K-Means Clustering

- Groups customers into a predefined number of clusters (K)
- It is simple and faster to execute
- It works well with numerical data such as Income and Spending score
- It is more scalable for large customer dataset

Hierarchical Clustering

- Creates clusters in tree-like hierarchy
- The number of clusters does not need to be fixed initially
- It is useful for understanding relationships between customers
- It is slower and less scalable for large datasets

Formula – Euclidean Distance :

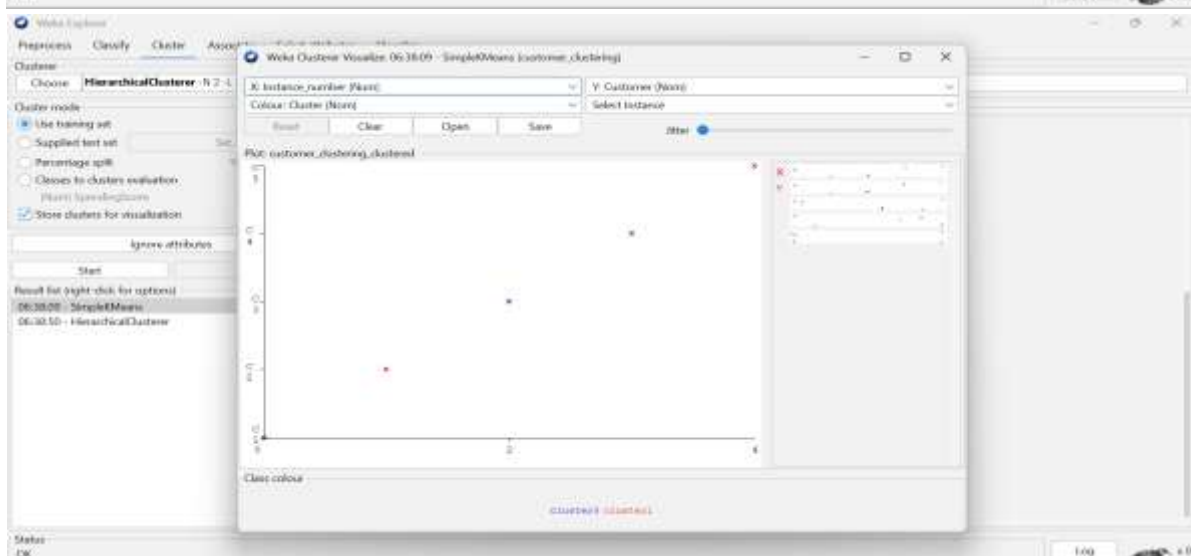
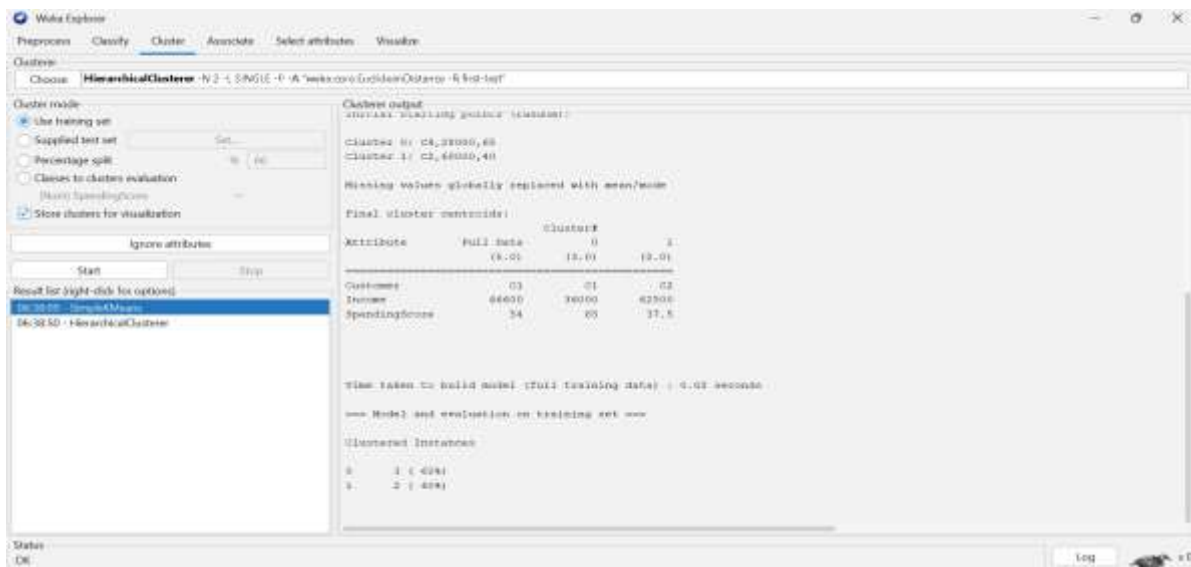
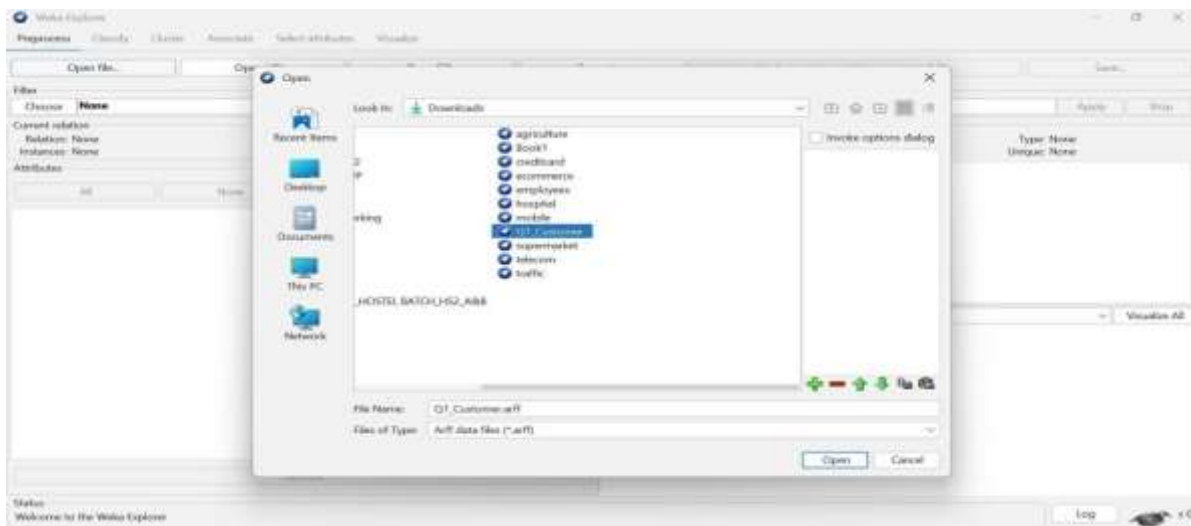
$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

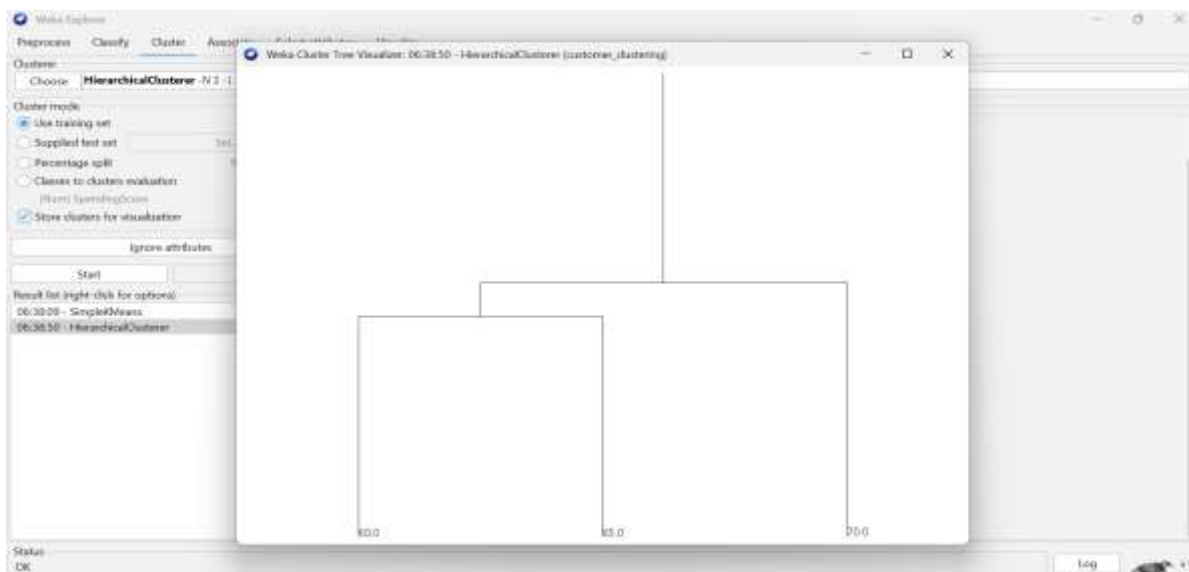
Analysis:

- In this dataset, customers can be grouped based on their income and spending score.
- C1 and C4 have lower income with higher spending, while C2 and C5 have higher income with lower spending.

Conclusion : K-means is better for large datasets customer data, while hierarchical clustering is useful for understanding

OUTPUT:





Q2. Dataset: Document Data

Document Word1 Word2 Word3

D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4 – 1 Mark)

Q2:

K-Means Vs EM Algorithms

K-Means :

- Assigns each document to only one cluster
- Works well when clusters are clearly separated.
- It may not handle overlapping documents effectively.
- It is simple and faster.

EM (Expectation-Maximization) :

- Assigns a probability of belonging to each cluster.
- Handles overlapping clusters better.
- It can represent uncertainty in document membership.
- It is more flexible but generally slower than K-Means.

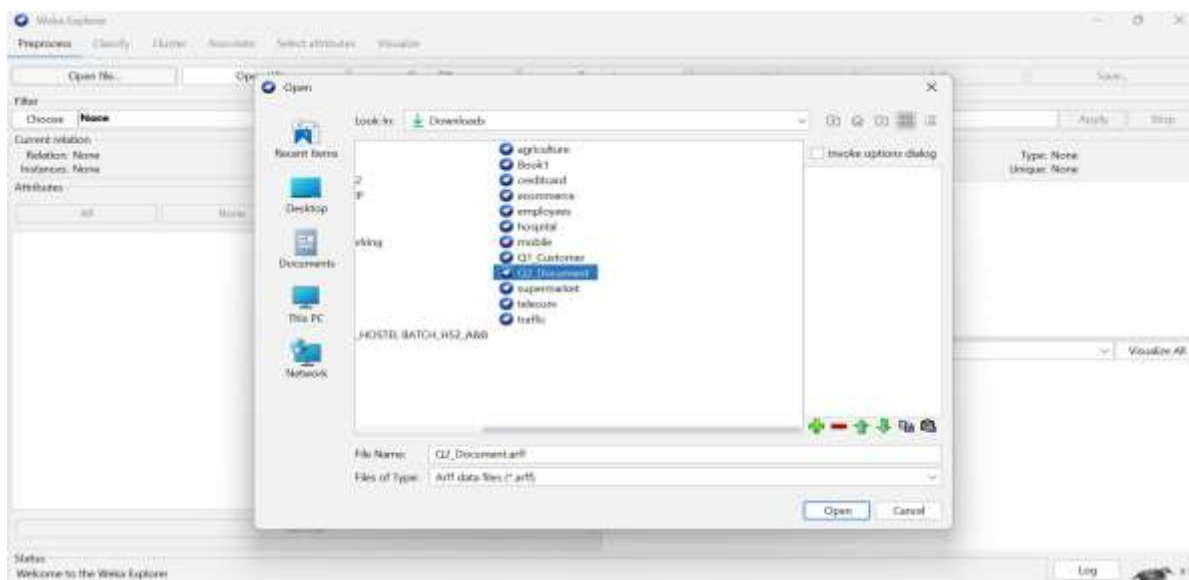
Analysis :-

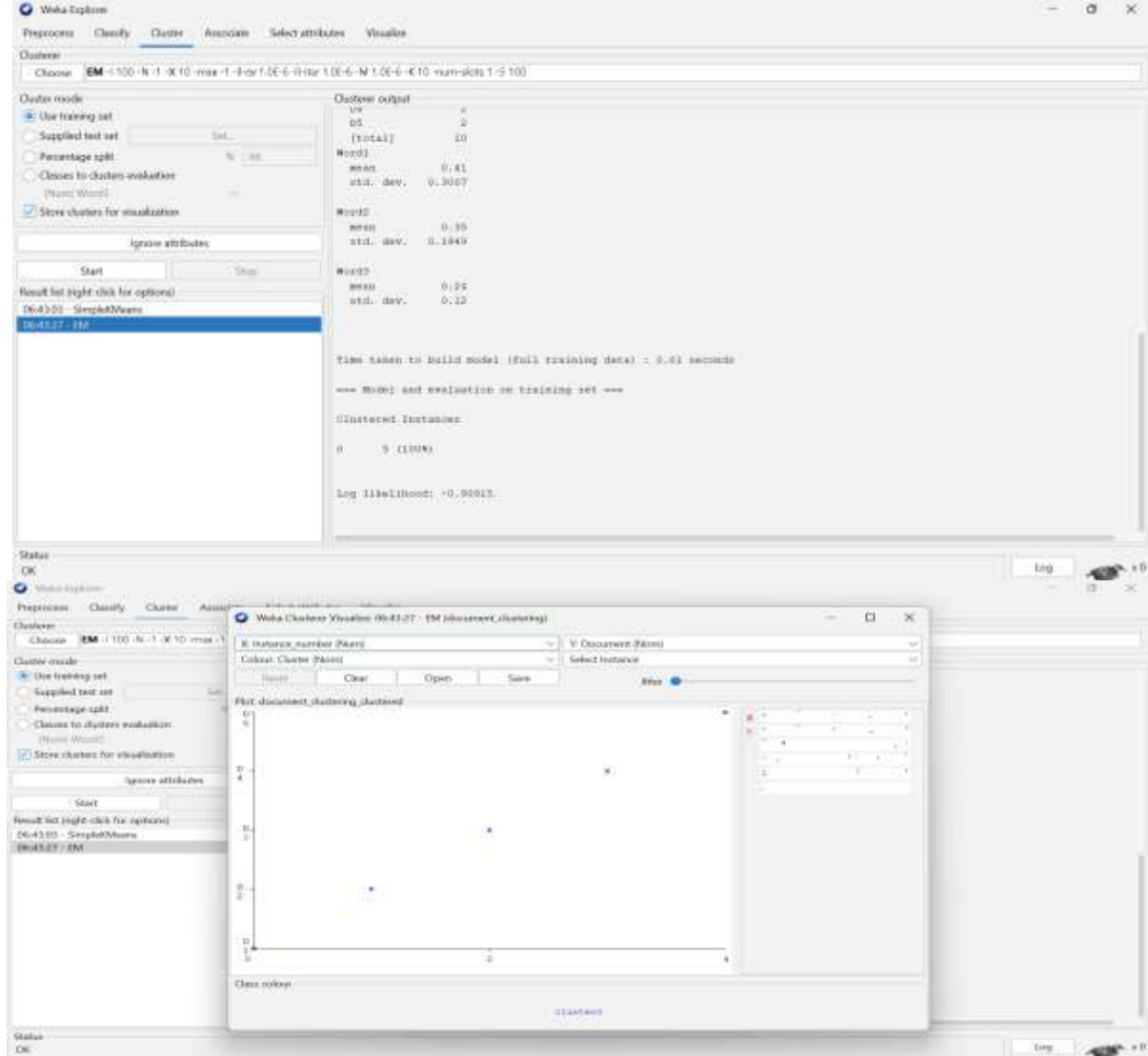
D1, D2 and D5 have similar word distributions, while D3 and D4 have a different distribution.

Conclusion :-

EM is more suitable for overlapping clusters because it provides probability-based assignments, while K-Means gives a hard assignment to a single cluster.

OUTPUT:





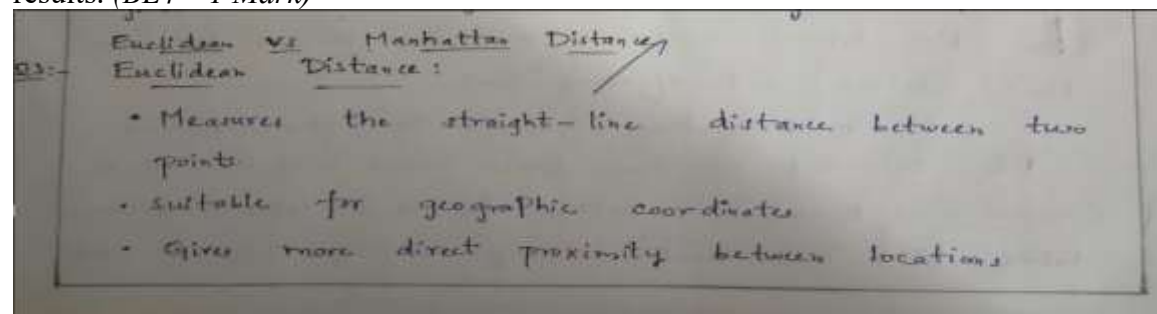
Q3. Dataset: Geographic Data

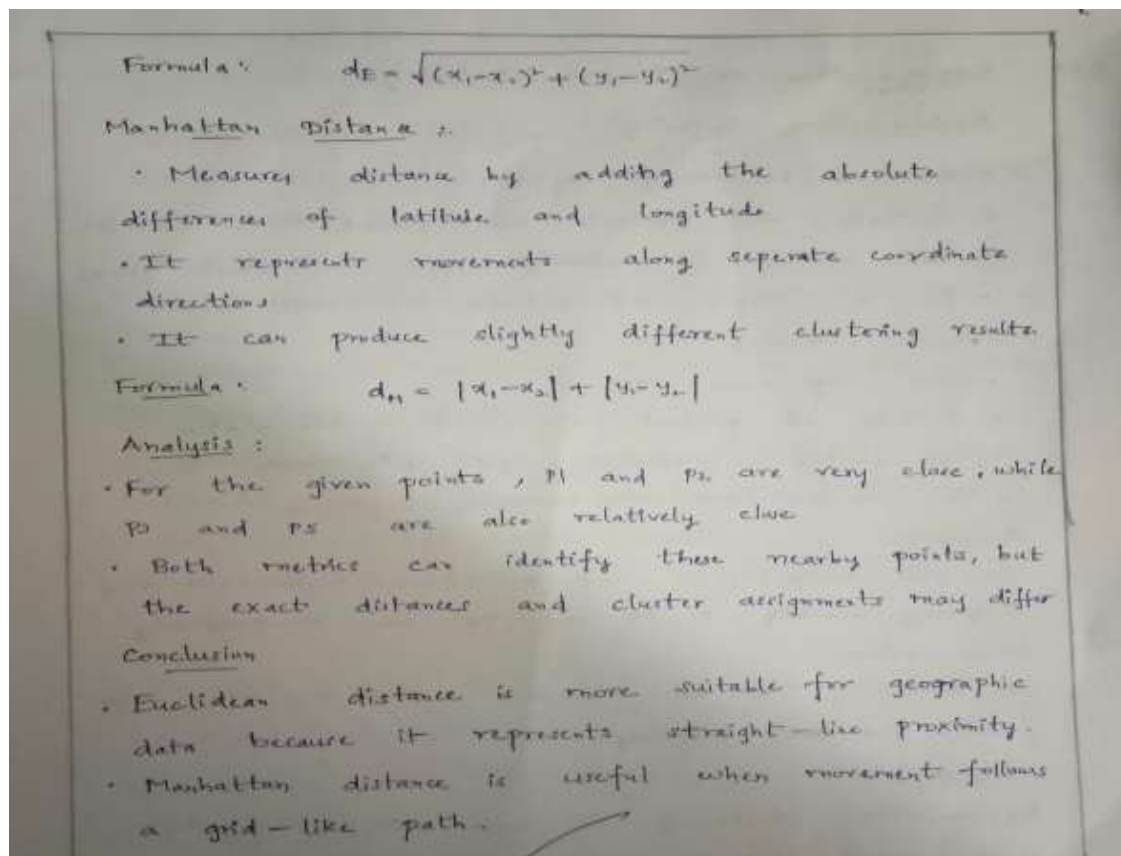
Point Latitude Longitude

P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

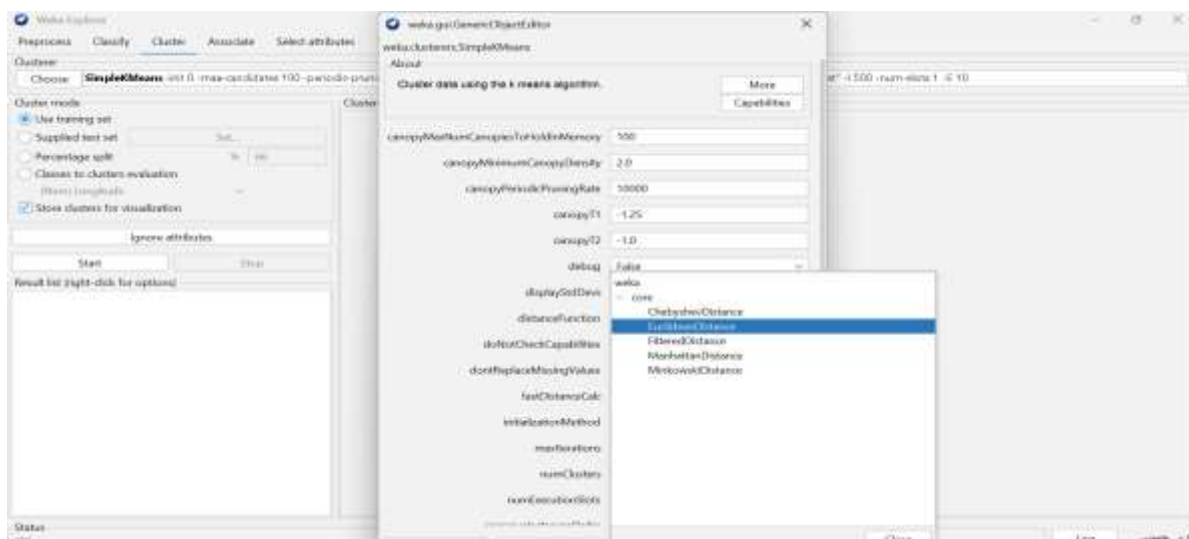
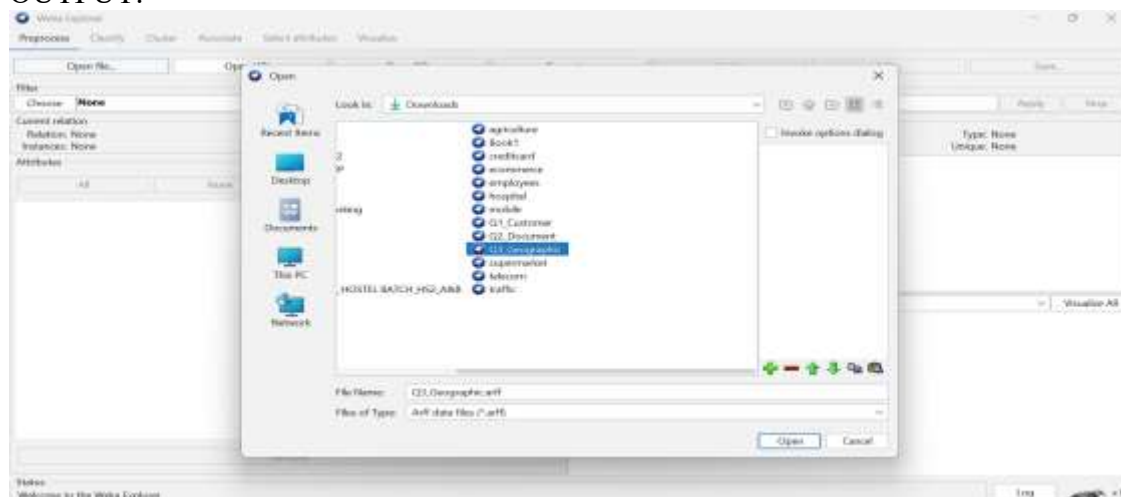
Question:

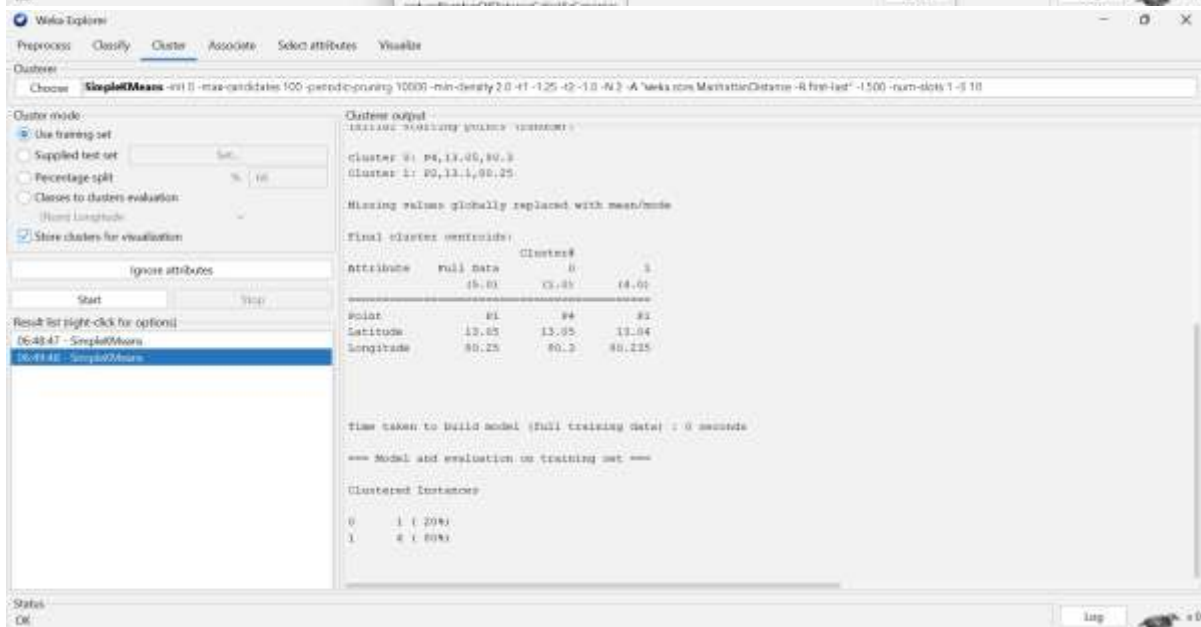
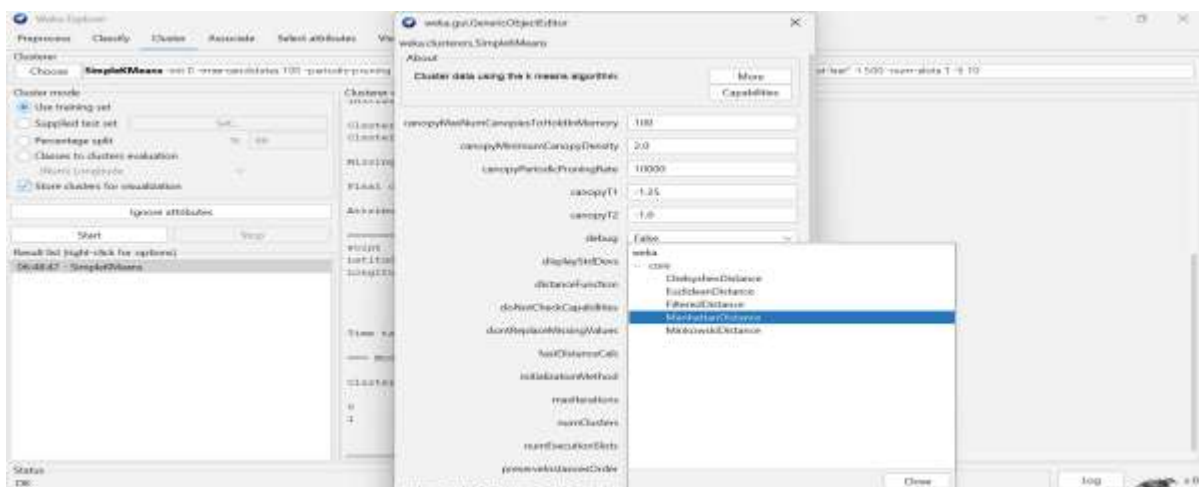
Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)

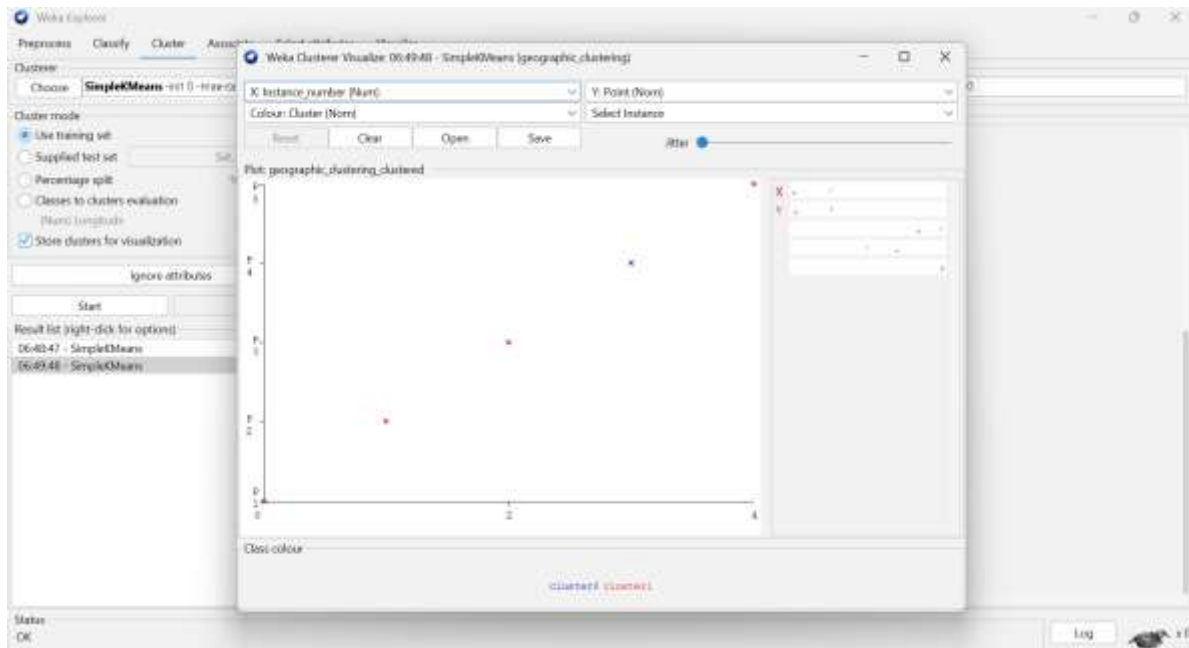




OUTPUT:







Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)

Q4: Agglomerative vs Divisive Clustering

Agglomerative clustering:

- Uses a bottom-up approach.
- Initially, all products are placed in one cluster.
- The cluster is gradually split into smaller groups.
- It is simple and suitable for small datasets.

Divisive clustering:

- Uses top-down approach.
- Initially, all products are placed in one cluster.
- The cluster is gradually split into smaller groups.
- It can be more computationally expensive.

Analysis:

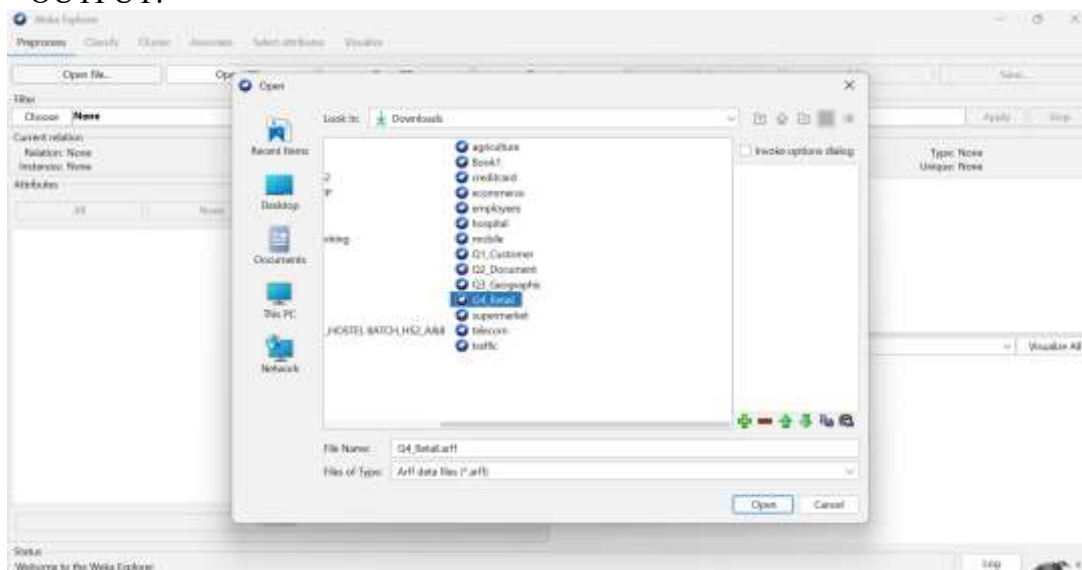
In this dataset, P1 and P3 have high sales and frequency, while P2, P4 and P5 have comparatively lower values.

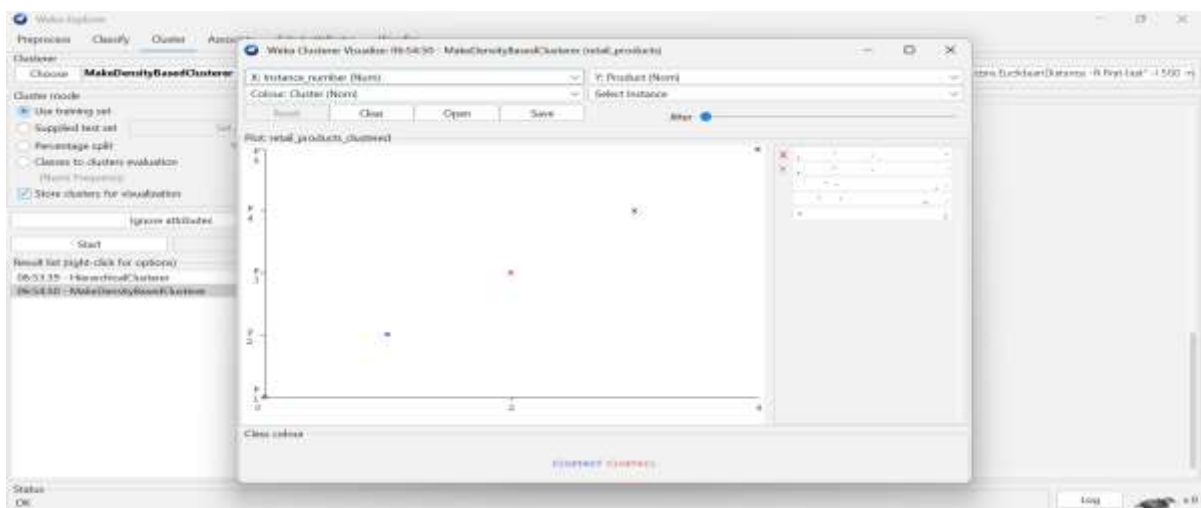
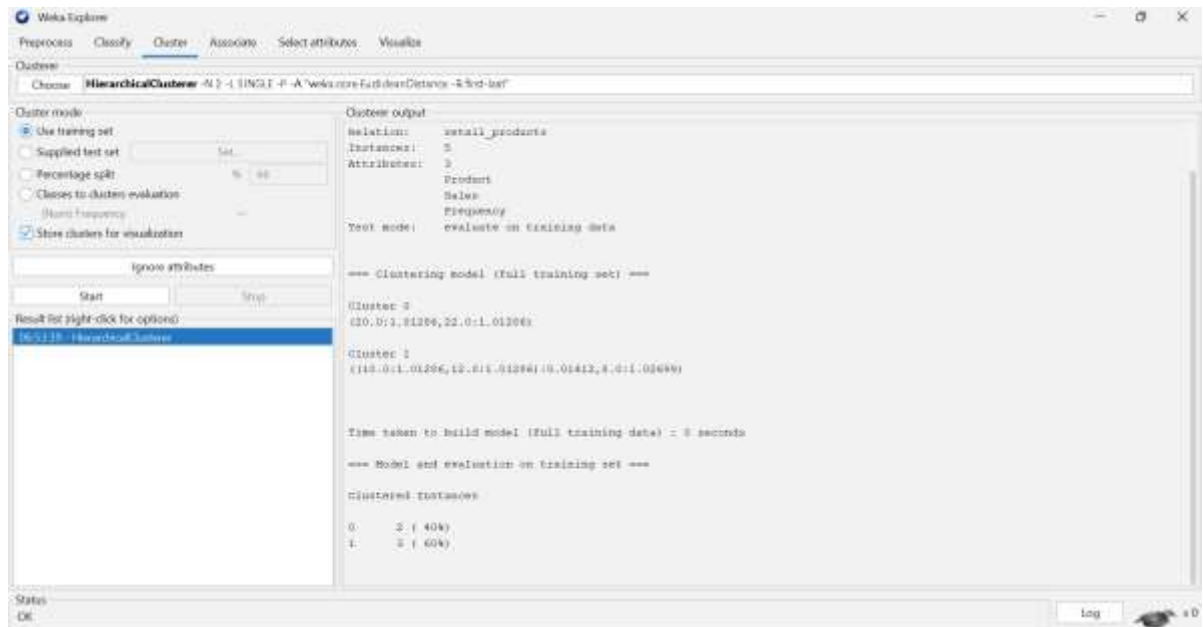
∴ products can be grouped based on their sales and purchase frequency.

Conclusion:

Agglomerative clustering is more suitable for this small retail dataset because it is simple, easy to implement, and can effectively group similar products.

OUTPUT:





Q5 . Dataset: Mixed Attributes

Item Price (₹) Category Rating

I1	500	A	4.5
I2	300	B	3.8

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster: Choose **SimpleKMeans** set 0 -max-iterations 100 -periodic-pruning 10000 -min-density 2.0 -t1 1.25 -t2 1.0 -N 2 -A "weka.com.ar.cluster.SimpleKMeans -R first-last" -1000 -max-iter 1 -S 10

Cluster mode: ☒ Use training set ☐ Supplied test set ☐ Percentage split ☐ Clones to clusters evaluation ☐ Normal Rating ☒ Store clusters for visualization

Ignore attributes:

Start Stop

Result list (right-click for options): 06:59:20 - SimpleKMeans

Cluster output

Cluster: 0: 350, 8, 3.5
Cluster: 1: 300, 8, 3.8

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster 0	Cluster 1
	(0.0)	(2.0)	(3.0)
Price	424	375	825.0000
Category	A	B	A
Rating	4.22	3.85	4.6

Time taken to build model (full training data): 0.01 seconds

== Model and evaluation on training set ==

Clustered Instances:

0	2 : 404
1	2 : 404

Status: OK

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster: Choose **MakeDensityBasedCluster** -W 100 -S 10 -W weka.com.ar.cluster.SimpleKMeans -t1 0 -max-iterations 100 -periodic-pruning 10000 -min-density 2.0 -t1 1.25 -t2 1.0 -N 2 -A "weka.com.ar.cluster.SimpleKMeans -R first-last" -1000 -max-iter 1 -S 10

Cluster mode: ☒ Use training set ☐ Supplied test set ☐ Percentage split ☐ Clones to clusters evaluation ☐ Normal Rating ☒ Store clusters for visualization

Ignore attributes:

Start Stop

Result list (right-click for options): 06:59:20 - SimpleKMeans
06:59:29 - MakeDensityBasedCluster

Cluster output

Normal Distribution: Mean = 3.14 StdDev = 2.0
Attribute: Category
Discrete Estimator: Counts = 1 3 (Total = 4)
Attribute: Rating
Normal Distribution: Mean = 3.45 StdDev = 0.15

Cluster: 1 Prior probability: 0.0714

Attribute: Price
Normal Distribution: Mean = 525.5833 StdDev = 20.548
Attribute: Category
Discrete Estimator: Counts = 4 1 (Total = 5)
Attribute: Rating
Normal Distribution: Mean = 4.6 StdDev = 0.0618

Time taken to build model (full training data): 8 seconds

== Model and evaluation on training set ==

Clustered Instances:

0	2 : 404
1	3 : 604

Log Likelihood: -4.6007

Status: OK

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster: Choose **MakeDensityBasedCluster**

Cluster mode: ☒ Use training set ☐ Supplied test set ☐ Percentage split ☐ Clones to clusters evaluation ☐ Normal Rating ☒ Store clusters for visualization

Ignore attributes:

Start Stop

Result list (right-click for options): 06:59:20 - SimpleKMeans
06:59:29 - MakeDensityBasedCluster

Weka Cluster Visualizer

X: Instance number (Num) Y: Price (Price)

Colour: Cluster (Nom) Select Instance

Plot: mixed_attributes-weka_files_unsupervised_attributes_remove-R1_clustered

Class colour

Status: OK